

Using Process Mining to Analyze Care Pathways in an Outpatient Oncology Clinic: A Case Study from a Brazilian Ambulatory Care Center

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Abstract. The complexity of oncology care pathways requires data-driven tools to enhance efficiency and quality. This paper reports a case study applying process mining to real-world workflows in a Brazilian outpatient oncology clinic. Using the PM² methodology, we modeled (i) departmental workflows for radiotherapy and chemotherapy and (ii) longitudinal care pathways from evaluation to discharge. The analysis highlighted common flows, process variants, and deviations from protocols. Comparisons across breast, prostate, and digestive cancers pathways revealed differences that can be relevant to care planning and resource allocation. Results demonstrate the value of process mining for actionable insights and continuous improvement in oncology care.

Keywords: Process mining, Process discovery, Cancer care unit, Care pathways, Patient care management.

1 Introduction

The increasing digitalization of clinical and operational workflows in hospitals has accelerated the adoption of advanced technologies designed to enhance both operational efficiency and the quality of patient care. In the oncology sector, the complexity of the patient journey presents significant challenges for the efficient management of resources. Traditional management approaches, grounded in personal judgment to assess operational workflows, frequently fall short in capturing the true complexity of process dynamics, even within computerized healthcare institutions.

In this context, the adoption of innovative technologies such as process mining has emerged as a promising approach for discovering, modeling, and optimizing processes based on real data extracted from hospital information systems [1]. Process mining enables the extraction, modeling, and analysis of actual processes using event logs generated by information systems, revealing nonconformities and bottlenecks in operational workflows. Studies show that its application in hospitals can significantly improve

patient flow management and resource allocation, resulting in greater efficiency and better quality of care [2].

In oncology care, the compliance and efficiency of clinical pathways become especially critical, as delays at the beginning or during treatment may negatively impact clinical outcomes [3]. Therefore, the analysis and continuous monitoring of clinical pathways throughout the patient journey are essential for enhancing both the quality of care and treatment outcomes.

In this context, process mining serves as a powerful tool to support such analysis, enabling managers and clinical teams to identify nonconformities, uncover bottlenecks, and improve care processes.

Thus, this paper presents a case study on the application of process mining to analyze care pathways in an outpatient oncology clinic. The analysis aimed to identify critical pathways and process variants that indicate inefficiencies or deviations from institutional workflows.

The insights gained from this analysis are intended to support the optimization of patient care pathways within the outpatient oncology setting.

The study was conducted at a Brazilian oncology clinic that treats all cancer types and offers chemotherapy, radiotherapy, clinical oncology, and supportive services. The clinic receives about 150 new cases per month and treats an average of 65 patients daily.

2 Related Works

Process mining has been applied in oncology to improve efficiency, compliance, and personalized care by discovering patient pathways and identifying bottlenecks. Its use in healthcare is well established, with several reviews covering both general [4–6] and oncology-specific domains [7]. According to Santos Garcia et al.’s [8] classification, most studies focus on process discovery (53%), followed by enhancement (19%), support (17%), and conformance checking (11%) [4]. Guzzo et al. [6] report similar results, noting discovery as the dominant approach.

The prevalence of process discovery likely reflects its role as the first step in any process mining initiative—understanding the process is essential before improvement or conformance checking. This aligns with our study’s objective: uncovering the actual workflows followed in daily practice. Another factor is the variability of clinical processes, which complicates adherence to protocols. As Guzzo et al. highlight, even patients with the same condition may follow different pathways depending on individual treatment responses.

Oncology was one of the first healthcare fields to adopt process mining. In 2008, Mans et al. [9] analyzed gynecological oncology patient flows, sharing our goal of discovering care pathways. Their study, however, focused only on gynecological tumors and used financial records, while ours covers all tumor types using clinical data from electronic medical records—allowing for more detailed workflow discovery. They also examined control flow, organizational, and performance perspectives, whereas we emphasize treatment flows across cancer types to support pathway improvements.

Pijnenborg et al. [10] applied process mining to palliative care for stomach and esophageal cancer, using data from the Netherlands Cancer Registry to compare pathways across hospitals. In contrast, our study centers on a single outpatient clinic, analyzing operational workflows across multiple cancers.

Savino et al. [11] investigated rectal cancer treatment at an Italian hospital, assessing adherence to European guidelines. Their approach—first discovering actual workflows, then analyzing deviations—inspires our strategy of mapping clinic processes before addressing bottlenecks and inefficiencies.

In Brazil, Iachecen et al. [12] studied lung cancer care within a health insurance provider, identifying diagnostic tests and treatment modalities. While their dataset spans multiple providers and populations, it is billing-focused and lacks clinical detail. Our study, though limited to one clinic, captures comprehensive care events, enabling more precise workflow analysis.

Despite these initiatives, few studies target outpatient oncology clinics specifically, highlighting the relevance of our work.

3 Methodology

This case study applied the PM² methodology, a widely used framework for process mining projects proposed by van Eck et al. [13]. PM² consists of six stages: planning, extraction, data processing, mining and analysis, evaluation, and process improvement (Figure 1). Its essence is to translate project goals into research questions that guide the analysis and generate actionable findings.

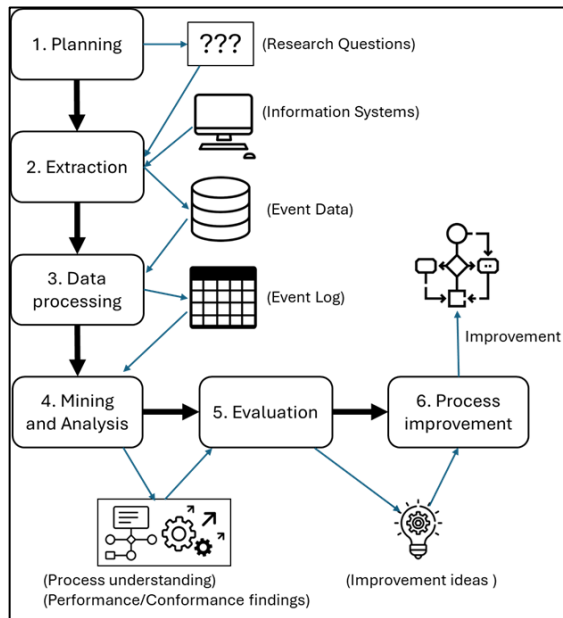


Fig. 1. PM² methodology overview. (Adapted from van Eck et al. [13]).

In our study, the first four stages were applied. During planning, research questions were defined from project goals. Extraction retrieved event data from clinical information systems, and data processing structured them into standardized event logs. Mining and analysis then applied process mining techniques to address the questions, revealing insights into performance, compliance, and patient flows.

Our objective was to understand outpatient treatment processes, identifying critical pathways and process variants that indicate inefficiencies or deviations from protocols (e.g., missed activities). Since the focus was on discovery and analysis, stages five and six—evaluation and process improvement—were left for future work.

3.1 Planning

The inputs to this phase are the organization's business processes, and the outputs consist of the research questions and the identification of the information systems from which event data can be extracted.

This case study considered two distinct business processes, both of which are relevant to the oncology patient's journey. The first concerns the patient's flow within the clinic during chemotherapy or radiotherapy sessions, which we will refer to as the Department Workflow. Efforts to improve this process focus primarily on its performance, aiming to reduce the patient's length of stay during these sessions. The research questions formulated for this process sought to investigate the critical paths; that is, those with the highest concentration of patients and, consequently, the greatest likelihood of bottlenecks and delays.

The second, referred to as Treatment Workflow, spans the entire treatment journey from initial multidisciplinary evaluation to discharge. Here, research questions aimed to investigate variations in patient flow within the clinic based on different types of cancer. Understanding these process variants can support more efficient planning of the resources involved in the care tasks associated with each treatment pathway.

3.2 Extracting

All events related to the business processes are stored in the clinic's management system, which encompasses everything from the electronic health record (EHR) to the billing module. This is a proprietary system developed by the clinic's own IT team, with Firebird as its database.

Event data were extracted through SQL scripts, which was validated and executed by the IT team. Developing these scripts required prior mapping of the database tables relevant to the event log, along with an analysis of the corresponding EHR modules and the data entry process. This ensured accurate event data selection.

Data transformation rules were embedded in the scripts to ensure data quality. The main rule verified the validity of event start and end dates, ensuring that the end date was later than the start date.

No personal data from patients or healthcare professionals were extracted. Only the information required to build the event log was retrieved: event identifier, executed

activity (clinical or administrative), start and end timestamps, scheduled date and time, patient arrival time, executor identifier, and resources used.

The extraction period was January–May 2025 for the Treatment Workflow and May 1–30, 2025 for the Department Workflow. No inclusion or exclusion criteria were applied beyond these timeframes. Extracted data were stored in a PostgreSQL database.

This study was approved by the Institutional Review Board (IRB) under number: CAAE: 75225723.7.0000.5497.

3.3 Data Processing

From the extracted event data, two event logs were created: one containing the events related to the department workflow and the other to the treatment workflow.

To enable more detailed analyses of the clinic's processes, additional columns were included in the event logs. The added data includes: the ICD-10 code, to identify the type of tumor; the first-line therapeutic approach (surgery, chemotherapy, radiotherapy, etc.); the patient's age group; and gender.

These event logs were stored in a PostgreSQL database and also exported to CSV format, which is one of the formats supported by the analysis tool

3.4 Mining and Analysis

The process analysis was conducted using Fluxicon Disco, chosen for three main reasons: it meets all project requirements, is user-friendly, and is easy to install without demanding significant hardware resources.

For this mining and analysis phase, a process discovery model was created to address each research question defined during planning. The identified process variants constitute the main findings of the analysis.

To manage the large number of process variants, a coverage threshold of 70% of cases was set. Thus, if a process has 100 variants but the top three represent 70% of cases, only these three are analyzed. For example, a patient may have an oncologist consultation, a nutritionist consultation, and a catheter cleaning on the same day. As the activity order does not affect the clinical care, the tool could discover six variants (3!) representing all possible sequences. Only the variants covering the largest number of cases, up to 70% of the total, are considered for analysis.

The individual analysis of the variants of the Department Workflow process was based on the observation of two key metrics: (i) case concentration, and (ii) average execution time per variant. These metrics were also evaluated across different data segments, including time periods, cancer types, age groups, and patient gender, to assess their stability and consistency for these different data samples.

The same analytical approach was applied to the Treatment Workflow process, except for the average execution time metric, which was excluded. This exclusion was due to the high variability in treatment duration, influenced by numerous factors such as therapeutic choices and individual patient responses.

Regarding the analysis of variants by cancer type, the investigation was limited to three cancer groups: breast, prostate, and digestive organ cancers, which together account for nearly 70% of the cases treated at the clinic

4 Results

The results are presented and discussed according to the stages of the PM² methodology, considering the first four stages, which define the scope of our case study.

4.1 Planning

The research questions were developed in collaboration with the clinic's clinical and operational management teams.

For the Department Workflow, the following questions were defined:

- What is the most common patient flow in the clinic's daily operations?
- What is the average process execution time?
- What is the percentage of significant deviations from the institutional workflow? (Significant deviations are considered those that should not occur; for example, a patient undergoing a procedure without prior registration at the front desk.)

For the Treatment Workflow, the following questions were formulated:

- What is the most common treatment pathway in the clinic?
- Are there significant differences among the treatment pathways for breast, prostate, and digestive organ cancers?

4.2 Extraction and Data Processing

Two SQL scripts were developed to extract event data from the clinic's management system: one for events related to the Department Workflow and another for events related to the Treatment Workflow. The first script resulted in a table with 13,400 rows, while the second extracted 67,469 rows.

These tables were exported as CSV files comprising the same columns of the database tables.

In addition to CSV files covering the entire period, we created five monthly files from January to May 2025. These multiple logs facilitate comparative analysis over time, as the tool can read and process several datasets simultaneously.

4.3 Mining and Analysis

A process model for the Department Workflow (January–May 2025) was generated to identify the clinic's most common patient flow (Figure 2). This model addresses the research question: "What is the most common patient flow in the clinic's daily operations?"

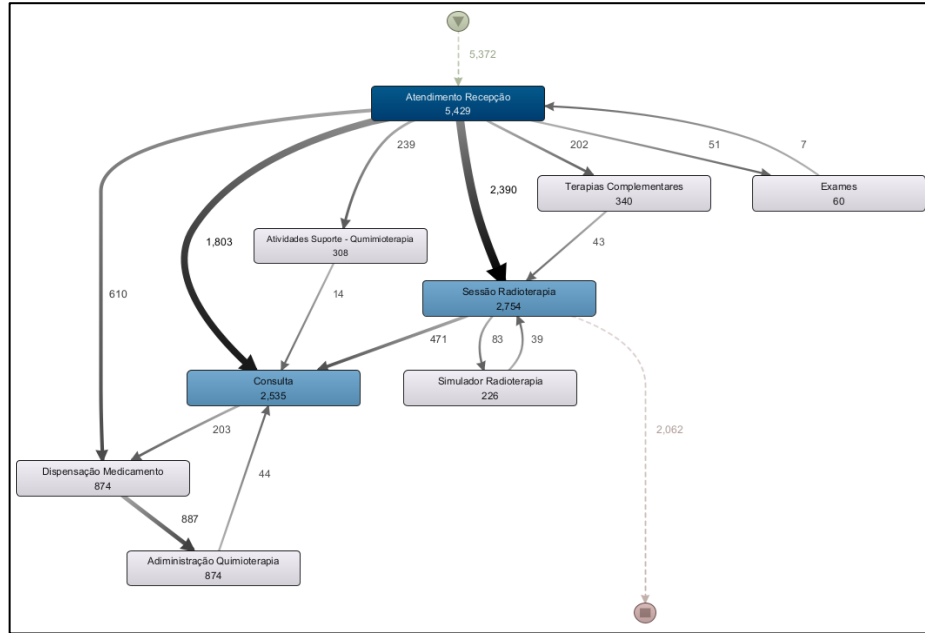


Fig. 2. Department process model – Jan 2025 to May 2025.

As shown in Figure 2, the most frequent activity flow corresponds to radiotherapy, with patients checking in at the front desk ("Atendimento Recepção") before the radiotherapy session ("Sessão Radioterapia"), totaling 2,754 occurrences. The second most common flow is evaluation consultations, where patients register at reception and proceed to a medical consultation ("Consulta"), with 2,535 occurrences. Chemotherapy infusion ranks third, occurring 874 times, and includes patient check-in, medication dispensing by the pharmacy ("Dispensação Medicamento"), and subsequent drug administration ("Administração Quimioterapia").

A detailed analysis of the process model in Figure 2 reveals several variant flows. For example, in the standard chemotherapy administration flow (reception → medication dispensing → chemotherapy), 610 occurrences followed the direct path, while 874 medication dispensing events were recorded. This indicates a variant in which some patients include a medical consultation between reception and medication dispensing.

Analyzing this initial model, which identifies the main process flows and the number of cases moving through each flow, provides valuable insights for the clinic's operational management. This information supports the right resource allocation to match patient volumes, helping reduce potential delays.

The process model in Figure 2 also addresses the research question: "What is the percentage of significant deviations from the institutional workflow?" A closer analysis reveals variants indicating significant deviations. For example, the process starts with 5,372 occurrences, yet there are 5,429 check-in records at reception desk, meaning 57 cases checked in more than once—approximately 1% deviation.

Another significant deviation can be seen in the medication dispensing activity. While there are 874 occurrences of dispensing, the model shows 887 outgoing transitions from this activity leading to drug administration. This also represents a small deviation, with a similar rate of approximately 1%.

A detailed investigation of these deviations is beyond the scope of this study. However, in this specific case, the most plausible hypothesis is that some medications were dispensed in separate batches—an accepted practice when two different lots of the same drug are used.

Another way to identify process deviations is by expanding the visualization of its variants. When displaying the process model showing 50% of variants (Figure 3), it is possible to detect another significant deviation from the institutional workflow: patients who did not check in at the reception desk and proceeded directly to medication dispensing (24 cases) or to a medical consultation (36 cases).

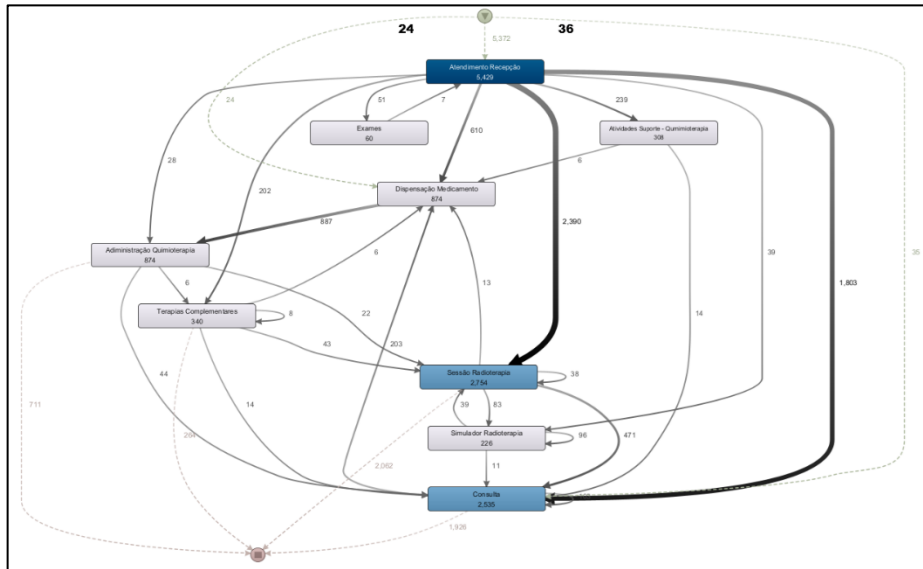


Fig. 3. Department process model showing the top 50% of variants.

Expanding the visualization of process variants enables clinic management to identify deviations from the institutional workflow and take corrective actions. It would be necessary to analyze the root causes and consequences of deviations to check if they affect efficiency or effectiveness of the process, however, it is beyond the scope of this study.

To address the research question “What is the average execution time of the process?”, we used the Global Statistics feature of Fluxicon Disco. The average execution time of the process shown in Figure 2 was 92.8 minutes, with 94% of cases completed in under 1 hour and 43 minutes. This aligns with the most frequent paths of the process — radiotherapy and medical consultations—typically completed in under an hour.

Chemotherapy administration, the longest workflow, ranges from 3 to 6 hours but accounts for only 16% of cases.

To conclude the Department Workflow analysis, different sample segments—period, cancer type, age group, and patient gender—were examined. Findings remained consistent across all segments, indicating that these variables do not significantly affect the clinic's operational workflow.

A process model for the Treatment Workflow (January–May 2025) was generated (Figure 4) to address the research question: "What is the most common treatment pathway in the clinic?" The model shows that the most frequent treatment in the clinic is radiotherapy, performed by 1,355 patients, followed by chemotherapy, applied to 490 cases.

The prevalence of radiotherapy as the most frequent treatment in the model can be explained by the clinic's role as a regional reference center for this service, receiving many patients who undergo oncological treatment at other institutions but attend this clinic specifically for radiotherapy.

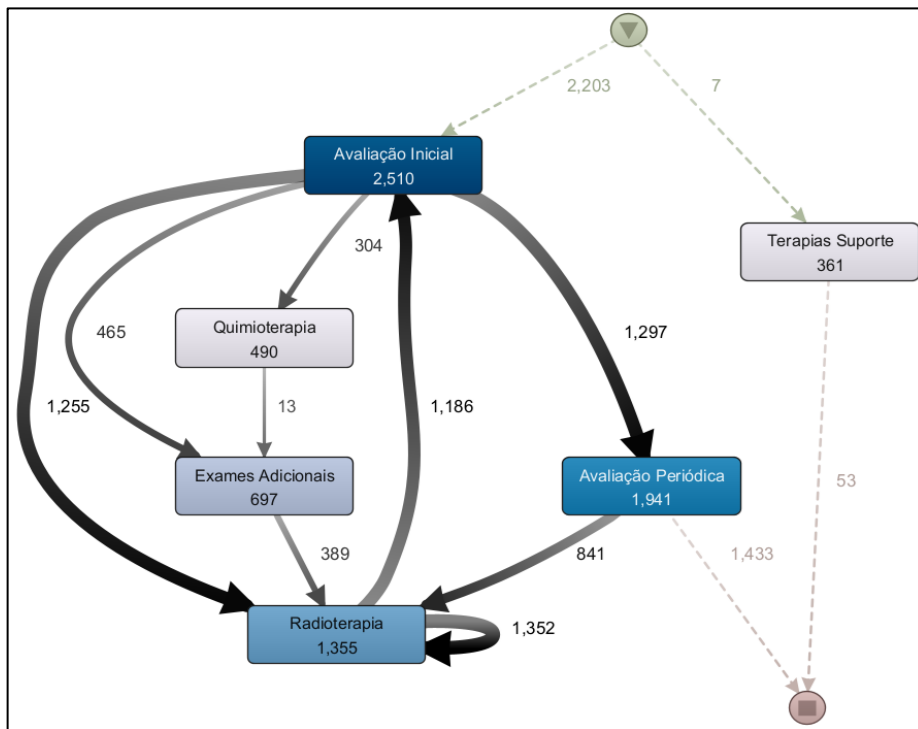


Fig. 4. Treatment process model – 2020 to 2025.

The second research question addressed to the Treatment Workflow model is: "Are there significant differences among the treatment pathways for breast, prostate, and digestive organ cancers?" The process model analysis for each of these cancer types indicates that there are. When comparing the treatment process model for prostate cancer

(Figure 5) with that for digestive organ cancers (Figure 6), it becomes clear that the former shows a more defined flow, with radiotherapy as the predominant treatment approach. In contrast, chemotherapy is more prevalent in the treatment of digestive organ cancers. Another noteworthy finding is the higher incidence of supportive therapy in cases of digestive organ tumors compared to prostate cancer cases.

Discovering treatment pathways for each cancer type provides valuable insights for both the clinic's operational management and quality teams. The management team can anticipate resource needs based on patient volume for each cancer type, while the quality team can assess whether treatment flows align with leading oncology guidelines.

Finally, different sample segments were analyzed for the Treatment Workflow, including period, age group, and patient gender. Findings remained consistent across all these segments, indicating that these variables do not significantly affect the treatment approach.

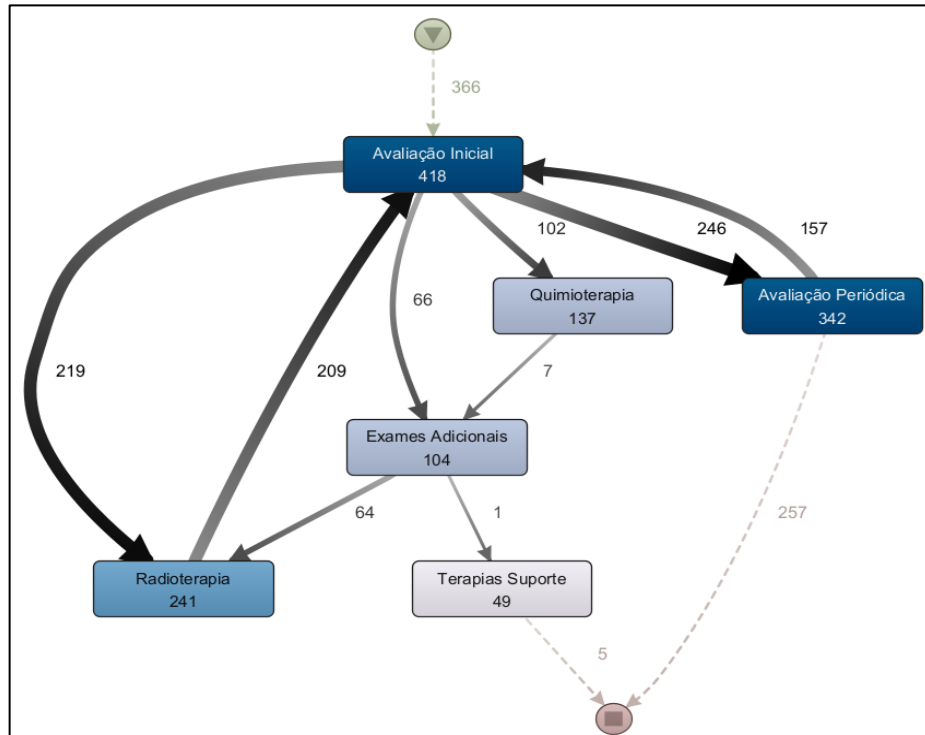


Fig. 6. Treatment process model – prostate cancer.

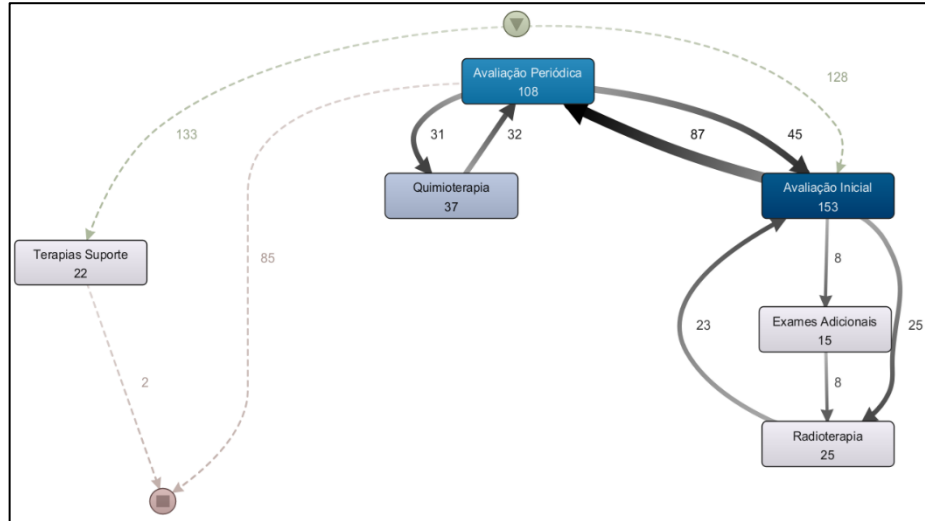


Fig. 6. Treatment process model – digestive organ cancer.

5 Conclusion

This study demonstrates the applicability and value of process mining techniques in uncovering and understanding the operational workflows of an outpatient oncology clinic. By applying the PM² methodology, we successfully mapped real-world treatment and departmental workflows, revealing the most frequent care pathways and identifying significant process variants and deviations.

Our findings show that radiotherapy and medical consultations are the most common daily activities in the oncology clinic analyzed in this case study, with chemotherapy following closely behind. Furthermore, the analysis of treatment pathways for breast, prostate, and digestive organ cancers revealed distinct patterns in care delivery, emphasizing the need for tailored operational strategies based on tumor type.

Although the current study focused primarily on process discovery and descriptive analysis, it lays the foundation for future conformance checking and process enhancement initiatives. These next steps may include identifying inefficiencies, refining workflows, and evaluating adherence to clinical guidelines—ultimately contributing to improved patient outcomes and more efficient resource allocation.

The results not only confirm the utility of process mining in oncology settings but also highlight its potential as a decision-support tool for healthcare administrators and clinical leaders.

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